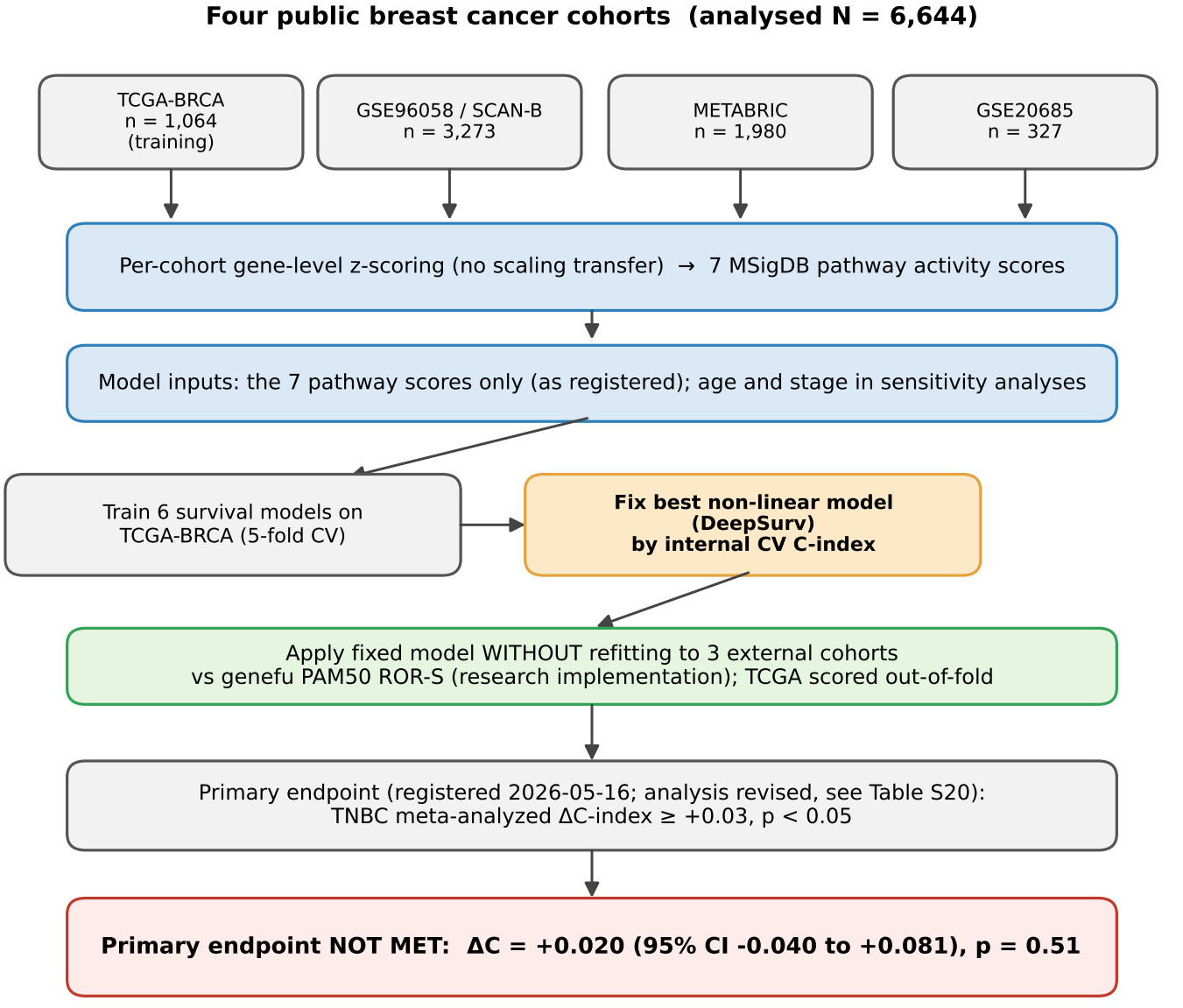
Four public breast cancer cohorts (analysed N = 6,644)
TCGA-BRCA
n = 1,064
(training)
GSE96058 / SCAN-B
n = 3,273
METABRIC
n = 1,980
GSE20685
n = 327
Per-cohort gene-level z-scoring (no scaling transfer) → 7 MSigDB pathway activity scores
Model inputs: the 7 pathway scores only (as registered); age and stage in sensitivity analyses
Train 6 survival models on
TCGA-BRCA (5-fold CV)
Fix best non-linear model
(DeepSurv)
by internal CV C-index
Apply fixed model WITHOUT refitting to 3 external cohorts
vs genefu PAM50 ROR-S (research implementation); TCGA scored out-of-fold
Primary endpoint (registered 2026-05-16; analysis revised, see Table S20):
TNBC meta-analyzed ΔC-index ≥ +0.03, p < 0.05
Primary endpoint NOT MET: ΔC = +0.020 (95% CI -0.040 to +0.081), p = 0.51